

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech. Branch : Petrochemical Engineering Semester : VI

Subject Code & Name: BTPCC604_Y21 Petrochemical Engineering-III

Max Marks: 60 Date:26/12/2024 Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.

2. Question No. 1 will be compulsory and include objective-type questions.

3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.

4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.

5. Use of non-programmable scientific calculators is allowed.

6. Assume suitable data wherever necessary and mention it clearly.

		(Level / CO)	Marks
Q.1	Objective type questions. (Compulsory Question)		
1	Molybdenum is used in many steel alloys to _____ a. to increase creep resistance b. to increase high temperature strength c. to increase resistance to scaling from oxidation d. none of these	Understand	
2	The selection of tube material in pipe still heater is on the basis of _____. a. corrosion characteristics of process fluid b. high temperature resistance of material c. tube metal temperature that governs creep resistance of the material d. all of these	Understand	1
3	As a thumb rule , Schedule number of pipe is an indicative of _____. a. diameter of pipe b. thickness of pipe c. the approx. max pressure that pipe can withstand d. all of the above	Understand	1
4	100 percent distilled line on superatmospheric phase diagram represent _____ condition of the stock. a. bubble point b. dew point c. both (a) and (b) d. none of the mentioned	Understand	1
5	Of the below mentioned methods for determining vaporization characteristic of given fraction, which one achieves least fractionation? a. ASTM distillation b. TBP distillation c. EFV distillation d. None of these	Understand	1
6	K value, equilibrium constant, more than one at given condition represent _____ distribution of that species in vapor phase than that in liquid phase. a. less b. more c. equal d. can't predict	Apply	1
7	Value of q = 1 for feed entering a distillation column indicates that feed is at its _____. a. dew point b. bubble point c. critical point d. none of these	Understand	1

8	Dehydration of natural gas or petroleum fraction like naptha and gasoline is carried out by using _____	Remember	1
	a. DEG b. MEA c. Lithum Chloride d. both (a) and (c)		
9	To increase smoke point of kerosene, the compounds removed in post treatment are _____	Understand	1
	a. olefins b. naphthenes c. aromatics d. paraffins		
10	Distillation column operating at lower value than that of minimum reflux ratio will require _____ number of stages for given separation.	Analyze	1
	a. infinite b. finite c. minium d. unpredictable		
11	Doctor process is used with reference to the post treatment of _____	Remember	1
	a. kerosene b. gasoline c. diesel d. LPG		
12	Which of these is not a inter-conversion method for vaporization characteristics determination	Understand	1
	a. Packie Method b. Underwood Method c. Edmistser Method d. Nelson Method		
Q.2	Solve the following.		12
A)	A multicomponent hydrocarbon mixture has the following composition on mole basis. i-butane - 0.20 , n-pentane - 0.25 and n-hexane – 0.55. Calculate dew point temperature and bubble point temperature for this mixture at 100 psia.	Apply	6
B)	In case of flash distillation, how do you calculate the flash? Derive the necessary expression.	Understand	6
Q.3	Solve the following.		12
A)	For a particular stock atmospheric ASTM-D158 distillation data is as follows: Volume % distilled 0 10 30 50 70 90 100 Temp °F 90 140 245 342 408 514 597 Obtain the atmospheric EFV data by using Van winkle method.	Apply	6
B)	Outline method for construction of phase diagram in super-atmospheric region with ASTM D-86 data available at the beginning.	Apply	6
Q.4	Solve Any Two of the following.		12
A)	Draw schematic representation of fractional distillation process. State the minimum essentials for the fractionation process.	Understand	6
B)	Write critically on : Pressure - An important design variable for distillation column.	Analyze	6

C)	Write an informative account on : Limits of operability of distillation column based on reflux ratio.	Analyze	6																									
Q.5	Solve Any Two of the following.		12																									
A)	Write an informative account on overhead corrosion in distillation unit.	Remember	6																									
B)	A quaternary mixture at its dew point composed of component a, b, c and d is to be fractionated to produce the products indicated. The system is to be considered essentially ideal. Calculate the minimum reflux by Underwood method. The reflux is a bubble point liquid.	Apply	6																									
	<table><tr><td>Component</td><td>x_f</td><td>α</td><td>x_D</td><td>x_B</td></tr><tr><td>a</td><td>0.047</td><td>4.2</td><td>0.1263</td><td>0.0000</td></tr><tr><td>b (LK)</td><td>0.072</td><td>1.57</td><td>0.1913</td><td>0.0010</td></tr><tr><td>c (HK)</td><td>0.600</td><td>1.0</td><td>0.6824</td><td>0.0350</td></tr><tr><td>d</td><td>0.281</td><td>0.8</td><td>0.0000</td><td>0.9649</td></tr></table>	Component	x_f	α	x_D	x_B	a	0.047	4.2	0.1263	0.0000	b (LK)	0.072	1.57	0.1913	0.0010	c (HK)	0.600	1.0	0.6824	0.0350	d	0.281	0.8	0.0000	0.9649		
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C)	Discuss the different aspect with respect to the design of vacuum distillation unit.	Understand	6																									
Q.6	Solve Any Two of the following.		12																									
A)	Why post treatment of different fractions is necessary? Briefly elaborate on chemical impurities.	Understand	6																									
B)	With neat sketch of box type of pipestill heater , give the general description of pipe still heaters.	Remember	6																									
C)	Discuss in details the different factors to be considered for design of radiant section of pipe still heater.	Analyze	6																									
	*** End ***																											